

Contemporary Mathematics

MATH 125 01 Fall 2026

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Christopher

Saginaw Valley State University

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Course Documents

Syllabus

Instructor Information

Instructor	Christopher Servant
Pronouns	he/him
Phone Number	(989) 964-4894
Email	cservant@svsu.edu
Office	SW 340
Student Hours	

Monday	1:00 PM - 2:00 PM
Tuesday	12:00 PM - 2:00 PM
Wednesday	1:00 PM - 2:00 PM

Note: Please do not contact your instructor via Canvas. Only use his email or phone above.

Note: If you need a more private appointment to discuss grades or another personal matter, additional student hours can be made to meet in my office or via ZOOM.

Course Information

Course Dates	August 31, 2026 - December 19, 2026
Meeting Times	M, W: 2:30 PM - 4:20 PM
Location	Brown Hall 107
Credits	4
Prerequisites	All Math Basic Skills or equivalent must be completed prior to taking this course. This is done by passing MATH 082 or MATH 072 with a C or higher.

Course Description

A liberal arts course providing an introduction to logical inference, statistics, probability, management science, social choice, concepts of size and shape, algorithmic processes and consumer mathematics. Emphasis on understanding rather than computational expertise.

Course Materials

Text *Math in Society* by David Lippman

You can view an online copy of this free, open source textbook at [Math in Society](#).

Online Course Component

In this class, you will have access to course notes, lecture videos, handouts, deadlines, and announcements on Canvas. If you are absent or class is canceled due to weather or other circumstances, it is your responsibility to log on and obtain missed notes, work, etc. Grades will be updated periodically on Canvas. It is also important to check your SVSU email regularly.

Course Outcomes

This General Education course introduces fundamental algebraic terminology and skills; demonstrates how ingenuity and creativity are essential in developing mathematical models to solve real world problems; cultivates critical thinking and logical reasoning for effective problem solving and rational decision-making in a complex technological society; and develops communication skills for translating real world situations into symbolic mathematical relationships and for interpreting mathematical results within their real world contexts.

The following lists the set of outcomes covered in this course:

1. Graphs and Shortest Path
2. Euler and Hamiltonian Circuits
3. Spanning Trees
4. Plurality
5. Runoff Voting and Borda Count
6. Copeland's Method and Approval Voting
7. Weighted Voting and Power
8. Hamilton's, Jefferson's, and Webster's Methods
9. Huntington-Hill and Lowndes' Methods
10. Logic and Statements
11. Truth Tables and De Morgan's Laws
12. Scheduling
13. Simple and Compound Interest
14. Annuities and Loans
15. Sets, Venn Diagrams, and Cardinality
16. Statistics, Data, and Sampling Methods
17. Presenting Data Graphically
18. Measuring Data

19. Probability

20. Fractals

Evaluation

Homework Each learning outcome in the course has one 10-point homework assignment in MyOpenMath. These assignments provide practice applying the concepts and skills associated with each outcome and help students prepare for quizzes and lab reports. A student's Homework score is the total number of points earned across all MyOpenMath homework assignments.

Quizzes Quizzes are administered approximately every fourth class meeting throughout most of the semester. Quizzes assess a student's ability to independently apply the concepts and skills developed through class activities, worksheets, and assigned course materials.

Each quiz is scored on a 0-3 scale:

- 3:** Demonstrates a strong understanding of the outcome with little to no error. Work is complete, mathematically correct, and clearly communicates the solution process.
- 2:** Demonstrates substantial understanding of the outcome but contains one or more minor errors. The overall solution process is appropriate and most of the work is correct.
- 1:** Demonstrates partial understanding of the outcome but contains significant errors or omissions. Some correct work is shown, but the student has not yet demonstrated competency.
- 0:** The outcome is not attempted or the work submitted does not demonstrate an understanding of the outcome.

Each quiz includes two in-class attempts. For each quiz, the highest score earned across all attempts is used when calculating the Quiz portion of the course grade.

A third attempt is available only after attending tutoring. Students must work with a tutor to review the relevant worksheet problems and their first two attempts before a third attempt will be granted.

Quizzes are not given as make-ups. The multiple-attempt structure is intended to accommodate absences and scheduling conflicts. Students who miss an in-class attempt may use a remaining attempt to demonstrate competency.

For the final quizzes of the course, only one in-class attempt may be provided during the final exam period. Additional attempts must be completed outside of class time by appointment.

Grading

Your final course grade will be determined by your performance in two categories: Homework and Quizzes. A letter grade will be determined separately for each category using the requirements below. Your final course grade will be the lowest of the two category grades.

Letter Grade	Homework Points	Quiz Points
A	180-200	51-60
B	160-179	42-50
C	140-159	33-41
D	120-139	24-32
F	0-119	0-23

Here are some examples showing how final course grades are determined:

1. A student earns 190 Homework Points and 56 Quiz Points. Both totals correspond to an A, so the student earns an *A* in the course.
2. A student earns 188 Homework Points and 47 Quiz Points. The Homework Points correspond to an A, while the Quiz Points correspond to a B. Since the lower of the two grades is a B, the student earns a *B* in the course.
3. A student earns 152 Homework Points and 54 Quiz Points. The Homework Points correspond to a C, while the Quiz Points correspond to an A. Since the lower of the two grades is a C, the student earns a *C* in the course.

Tips for Success

1. Read the textbook before attending class. The instructors explanations will make more sense and when they don't you will be more likely able to ask a relevant question.
2. Attend each class session, both physically and mentally.
3. Read the syllabus and look over the schedule on the last page of the syllabus.
4. Build community with other students in the course and create study groups.
5. Work outside of class. Math is learned by doing not seeing.
6. Ask for help. Stop in to see me, email me, call me, or ask others in class for help.
7. For free math tutoring; find information about the Math & Physics Learning Center at [MPLC](#)

Academic Integrity Policy

In MATH 132B, any occurrence of cheating or academic dishonesty may be officially documented. The student or students involved may then receive a copy of the documentation. The first occurrence may result in failure of the coursework and will be reported to the Office of Student Conduct Programs for the record. Any additional occurrence will be reported to the Office of Student Conduct Programs for adjudication.

You may work in groups or with tutors on homework and lab reports when permitted, but you must complete your own work. Quizzes must be completed independently without resources. Inappropriate use of laptops, cell phones, calculators, tablets, websites, outside assistance, or other resources is considered cheating. If you are unsure whether something is permitted, ask your instructor before using it.

See [SVSU's Academic Integrity Policy](#) for additional information.

Withdrawal Policy

Please consult an academic advisor and the Student Financial Services Center before withdrawing from any course. They can help you understand the possible effects that withdrawing may have on your degree progress and financial aid.

See [SVSU's withdrawal information](#) for additional details.

Important Dates

- September 4, 2026, is the last day to withdraw with a 100% refund and have the course deleted.
- September 18, 2026, is the last day to withdraw with a 100% refund and a “W” grade.
- October 2, 2026, is the last day to withdraw with a 50% refund and a “W” grade.
- November 13, 2026, is the last day to withdraw with a “W” grade.

Disability Statement

Students with disabilities that may restrict their full participation in course activities are encouraged to meet with the instructor or contact the SVSU Office of Accessibility Resources and Accommodations, located in Wickes 260, at 964-7000 for assistance.

Non-Discrimination Statement

SVSU does not discriminate based on race, religion, color, gender, sexual orientation, national origin, age, physical impairment, disability, or veteran status in the provision of education, employment, and other services.

Land Acknowledgment Statement

At Saginaw Valley State University, we recognize that we are located on the ancestral homelands of the Anishinaabe, Wyandot, and Sauk. We specifically acknowledge the Saginaw Chippewa Tribe and the land ceded in the Saginaw Treaty of 1819, on which we reside. We appreciate the importance of preserving and honoring the traditions of Indigenous People who are an integral part of our shared history. As we pledge to serve as responsible stewards of the land, we endeavor for this history to inform our teaching, scholarship, and commitment to community.

Tentative Syllabus

This syllabus and course schedule are subject to change if class needs warrant changes due to illness, inclement weather, or other circumstances.

Notes

Notes: Graphs Vocabulary and Representations

Objectives

- Translate a real situation into a graph.
- Identify vertices, edges, loops, degrees, paths, circuits, and connected graphs.
- Explain why the drawing of a graph may change while its connections remain the same.

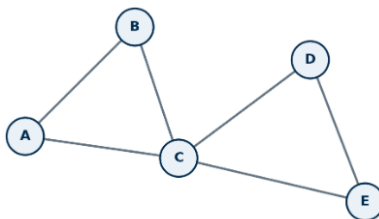
Core Vocabulary

Definition 1	Graph.	◆
Definition 2	Vertex.	◆
Definition 3	Edge.	◆
Definition 4	Loop.	◆
Definition 5	Degree of a Vertex.	◆
Definition 6	Path.	◆

Definition 7	Circuit.	◆
Definition 8	Connected Graph.	◆
Definition 9	Weight.	◆

Reading a Graph

Example 10 Reading a Graph. Use the graph below to answer the questions.



1. How many vertices and edges are present?
2. Find the degree of each vertex.
3. Give one path from A to E.
4. Give one circuit that starts and ends at C.
5. Is the graph connected? Explain.
6. Are there any loops? If not add one.



Check Your Understanding

1. Two edges cross in a drawing, but no dot appears at the crossing. Is the crossing a vertex? Explain.

2. Can two different-looking drawings represent the same graph? What must remain unchanged?

3. Can a connected graph contain a vertex of degree zero? Explain.
